Ayush Shivkumar

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Indian Institute of Science Education and Research (IISER) Thiruvananthapuram, Kerala, India

Education

- **BS–MS Dual Degree in Physics (Minor in Data Science)** *2021 – 2026*
  Indian Institute of Science Education and Research (IISER) Thiruvananthapuram Kerala, India
  - Integrated Bachelor's + Master's in Physics
  - **CGPA: 8.89/10.0**
- **Secondary & Higher Secondary Education (A-level equivalent)** *- 2021*
  Indian School Al Wadi Al Kabir (Oman) & RJ Thakur Junior College of Science (India) Oman & India
  - Grades: 90% (Secondary), 96.8% (Higher Secondary)

Research and Technical Experience

- **Master's Thesis on Protoplanetary Disks** *2025/08 – Present*
  IISER Thiruvananthapuram | Supervisor: Dr. Manash Samal (PRL Ahmedabad)
  - Analyzing multi-epoch spectroscopy (ESPaDOnS, X-shooter, Keck/NIRSPEC) and JWST MIRI MRS data of HD 163296 to probe disk structure, gas excitation, and accretion variability.
  - Investigating the interplay between gas dynamics, stellar irradiation, and disk substructure through multiline molecular emission analyses and slab modelling.
  - Developing skills in combining multi-wavelength datasets for detailed physical and chemical characterisation of protoplanetary disks.
- **Project on Protostellar Outflows & Gas Kinematics** *2024/05 – 2024/11*
  Physical Research Laboratory (PRL) Ahmedabad, Dept. of Space, Govt. of India | Guide: Dr. Manash Samal
  - Analysed clean ALMA data of multi-outflow systems.
  - Identified compact structures likely hosting protostellar sources.
  - Derived physical parameters of compact sources and calculated SiO jet properties from spectral line analysis.
- **Summer Internship on Young Star-Forming Regions** *2023/06 – 2023/08*
  Indian Institute of Space Science and Technology, Dept. of Space, Govt. of India | Guide: Prof. Sarita Vig
  - Detected and classified young stellar objects (YSOs) into Classes I, II, and III using Spitzer and 2MASS data.
  - Developed Python scripts for SED fitting using LMFIT; derived dust temperature and column density maps using Herschel PACS and SPIRE data.
- **Band E_{peak} and α Correlation in Optically Thin GRBs** *2025/01 – 2025/11*
  IISER Thiruvananthapuram | Supervisor: Dr. Shabnam Iyyani
  - I developed a novel simulation framework for optically thin GRB prompt spectra as part of my minor degree project in data science.
  - Performed joint likelihood fitting with the Band model, and investigated correlations between low-energy spectral index α and peak energy E_{peak}.

Publications

1) Bordoloi, P. P., **Shivkumar, A.**, & Iyyani, S. (2026). *E_{peak}–α Correlation in Time-Resolved GRB Spectra: A Bottom-Up Approach with Optically Thin Inverse Compton Scattering Model*. arXiv:2606.04894.

RESEARCH INTERESTS

Star and planet formation, protoplanetary disks, protostellar evolution and outflows, disk chemistry and gas dynamics, accretion variability, multi-wavelength observational astronomy and spectroscopy

TECHNICAL SKILLS

- **Languages:** English - IELTS Overall Band = **8.5** (C2 level) (**L = 9.0, R = 9.0, W = 8.0, S = 8.0**); TOEFL-equivalent = 115-117. Fluent in Tamil and Hindi with basic understanding of French.
- **Programming:** Python (advanced), LaTeX
- **Astronomy Techniques:** Spectral line analysis, continuum-fitting, and modelling, Photometric calibration, Timeseries Analysis & Period Search (GLS, PDM, Continuous Wavelet analysis), SED construction, joint likelihood fitting (3ML), Multi-component Gaussian fitting, FITS spectra and data cubes, ML & Python automation
- **Software/Astronomy-Specific Packages:** Jupyter, TOPCAT, SAOds9, VOSA, Astrodendro, SpectralCube, iris, Lightkurve, PyAstronomy, specutils, 3ML, LMFIT, PySTELLA, JWST Calibration Pipeline, bypaselines

ACADEMIC AWARDS AND ACHIEVEMENTS

- **INSPIRE-SHE Scholarship** *2021-2026*
  Awarded by the Department of Science & Technology, Government of India
  - Awarded to top 1% of students nationwide based on academic excellence in science. A highly competitive merit scholarship of ~USD $4,400 total funding over 5 years.
- **ZTF Summer School 2024 (Online Participant)**
  Zwicky Transient Facility Partnered with the University of Minnesota, Twin Cities 2024/08
  - Selected as an online candidate (1 of only 70 worldwide) to work with ML algorithms for automated detection and classification of transient phenomena. The selection process involved academic and research experience, as well as a Python-based assignment.

LEADERSHIP AND ORGANIZATION EXPERIENCE

- Built a radio telescope prototype using repurposed materials and a simple gravitational lensing demonstration as part of the annual science fest of IISER Thiruvananthapuram, ANVESHA.
- I have participated in outreach events such as telescope sessions and organising space-themed quizzes with an emphasis on promoting astronomy and astrophysics as part of the student-run astronomy club. I played an active role in writing blog posts and social media content for the Physics and Astronomy clubs, and an article I wrote on molecular clouds was also featured in the Chemistry Club's annual magazine: *Synthesis*

REFERENCES

1. **Dr. Manash Samal**
   Associate Professor, Astronomy and Astrophysics Department
   Physical Research Laboratory, Ahmedabad, India
   Email: manash@prl.res.in
   Phone: +91-79-26314613
   Relationship: Master's Thesis Advisor, Summer Project Guide
2. **Dr. Shabnam Iyyani**
   Assistant Professor, School of Physics
   Indian Institute of Science Education and Research Thiruvananthapuram, India
   Email: shabnam@iisertvm.ac.in
   Phone: +91 (0)471 - 2778304
   Relationship: Minor Degree Project Supervisor, Course Instructor